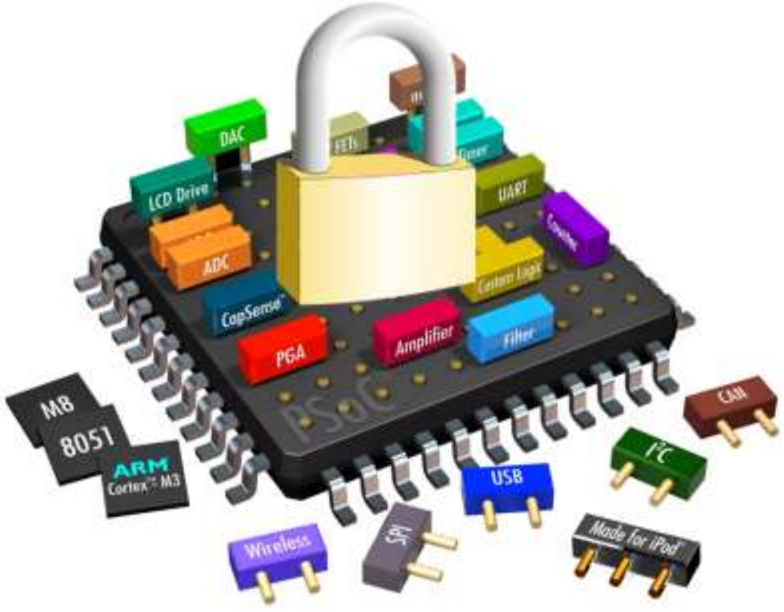
Hardware Security

TEE Examples: Intel TDX



What is Intel TDX?

- Intel Trust Domain Extensions (TDX) is an architectural extension in 4th Generation Intel Xeon Scalable Processors that enables confidential computing.
- TDX allows deployment of virtual machines in Secure-Arbitration Mode (SEAM) with encrypted CPU state and memory, integrity protection, and remote attestation.
- TDX enforces hardware-assisted isolation for virtual machines and minimizes the attack surface exposed to host platforms, which are considered untrustworthy in confidential computing's threat model.

Core Security Principles

01

Memory Confidentiality

TD (Trust Domain) data is encrypted when offloaded to main memory using TD-specific cryptographic keys known only to the processor

02

CPU State Confidentiality

Virtual CPU states are protected during context switches, stored in encrypted metadata

03

Execution Integrity

Protects TD execution from host interference, detecting malicious changes in virtual CPU states

04

I/O Protection

Peripheral devices cannot access TD private memory; shared memory requires explicit TD consent

TDX Threat Model

- Adversary Capabilities
  - Physical or remote access to computer
  - Control over boot firmware, SMM, host OS, hypervisor
  - Control of peripheral devices
  - Ability to interrupt TDs and manipulate memory
  - Physical and hardware attacks on buses
- What TDX Protects
  - TDX prevents adversaries from compromising confidentiality and integrity of TD memory and CPU state.
  - However, it cannot guarantee availability - adversaries can launch DoS attacks by controlling compute resources.

Trusted Computing Base (TCB)

Hardware Components

TDX-enabled Intel processors with VT, MKTME, and SGX technologies

Intel-Signed Software

TDX Module, NP/P-SEAM Loaders, and architectural SGX enclaves for attestation

Tenant Software

Software stacks running within TDs, owned and controlled by the tenant

Trust Requirements

Tenants must trust Intel for development, manufacturing, and signing of components, plus Intel's Provisioning Certification Service

Building Block: Intel VT

- Virtualization Technology
  - Intel VT provides hardware-assisted virtualization with better performance, isolation, and security than software-based approaches.
    - VMX root mode for hypervisor
    - VMX non-root mode for guest VMs
    - Extended Page Table (EPT) for address translation
    - VT-d for I/O device isolation
- TDX Integration
  - TDX extends VT with new SEAM VMX modes.
  - Untrusted hypervisor → TDX Module
  - VMX root mode → SEAM VMX root mode
  - VMX non-root mode → SEAM VMX non-root mode

Building Block: Intel MKTME

1

TME Foundation

Total Memory Encryption encrypts entire memory using single transient key generated at boot-time with AES-XTS algorithm

2

MKTME Extension

Multi-key TME supports multiple keys for page-granularity encryption using Host Key Identifiers (HKIDs)

3

TDX Application

TDX Module controls memory encryption for TDs, partitioning HKID space into private and shared ranges

Building Block: Intel SGX

- SGX protects against memory bus snooping and cold boot attacks by partitioning applications into secure enclaves. Memory encryption protects enclave contents from unauthorized access.
  - Enclave Page Cache (EPC) for secure memory
  - Memory Encryption Engine (MEE)
  - Local and remote attestation
  - 18 new ISA instructions
- TDX Integration
  - TDX leverages SGX's remote attestation mechanism.
  - The Quoting Enclave verifies and signs TD reports, enabling remote verification of TD authenticity.